

# Portfolio Optimization & Walk-Forward Backtesting Platform

## Technical Report

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### 1. Executive Summary

This project is an end-to-end quantitative research platform for building, comparing, and rigorously validating equity portfolio allocation strategies. Rather than producing a single "optimal" portfolio, it treats allocation strategy selection as an empirical question: five competing construction methodologies are built on identical data, evaluated in-sample, re-optimized on a rolling basis and tested strictly out-of-sample, stress-tested against parameter and market-regime variation, and finally decomposed to explain *why* each strategy performed the way it did.

The codebase is organized as a sequential pipeline of numbered "phases," moving from raw data acquisition through to daily-level performance attribution. It is designed to answer the question a portfolio manager actually needs answered before committing capital: *not just "which strategy has the best backtest Sharpe," but "does that edge survive realistic transaction costs, out-of-sample re-estimation, and different market regimes?"*

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### 2. Objective & Scope

**Primary objective:** Compare five portfolio construction methodologies on a common equity universe and determine which one delivers superior risk-adjusted performance once realistic frictions and out-of-sample validation are applied; not just in a single backtest, but robustly across time and parameter choices.

**In scope:**

- Portfolio construction across five allocation models
- Discrete, cost-aware execution modeling
- Point-in-time (no-lookahead) rolling re-optimization and walk-forward testing
- Parameter sensitivity and historical stress-period robustness testing
- Daily-level performance attribution and risk decomposition

**Out of scope:** live trading/execution, intraday data, options or derivatives overlays, and non-equity asset classes.

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### 3. Data Universe & Acquisition

| Component    | Detail                                                                                                                                                                                 |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Universe     | 50 large-cap U.S. equities across Technology, Financials, Healthcare, Industrials, Consumer Staples, Energy, and Materials                                                             |
| Benchmark    | SPY (SPDR S&P 500 ETF Trust)                                                                                                                                                           |
| Market index | ^GSPC (S&P 500 Index), used for Black-Litterman risk-aversion / equilibrium-return calibration                                                                                         |
| History      | 2000-01-01 to 2026-09-01 (~26.7 years)                                                                                                                                                 |
| Source       | yfinance — bulk download() for daily OHLCV/Close prices, per-ticker Ticker.history() / get_shares_full() for shares outstanding (used to reconstruct historical market capitalization) |
| Fields used  | Adjusted close prices, daily returns, shares outstanding (for market-cap-weighted priors)                                                                                              |

The 50-name universe spans a deliberately diverse sector mix (e.g., JPM, GS, MS in Financials; JNJ, MRK, ABT, GILD in Healthcare; XOM, CVX in Energy; INTC, QCOM, AMD-adjacent semis) so that the optimization models have meaningful diversification and correlation structure to work with, rather than a sector-concentrated universe that would make every model converge to similar weights.

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## 4. Data Engineering & Preprocessing

Before any optimization runs, the pipeline performs:

- **Missing data handling:** forward-fill then back-fill of gaps, followed by a hard drop of any residual missing rows, so every ticker has a fully aligned daily price series.
- **Duplicate-index removal** and chronological sorting of all price and market-cap panels.
- **Historical market capitalization reconstruction:** since yfinance does not provide historical market cap directly, it is reconstructed as price  $\times$  shares outstanding at each historical date, using per-ticker share-count history.
- **Timezone normalization:** a subtle but important fix applied during development — yfinance's bulk download() endpoint (used for prices) returns a timezone-**naive** index, while Ticker.history() (used for share counts) returns a timezone-**aware** index localized to the exchange timezone. Left unreconciled, this caused every timestamp comparison against the market-cap panel to fail. The market-cap index is now explicitly stripped of timezone information immediately after construction, so all downstream date comparisons operate on a single consistent (naive) timestamp convention.

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## 5. Portfolio Construction Methodologies

Five allocation strategies are built on the same 50-asset universe and directly compared:

1. **Equal-Weight** — a 1/N baseline; the standard null hypothesis that "smart" allocation should beat.
2. **Simulated Max-Sharpe (Monte Carlo)** — 1,000,000 randomly generated long-only weight vectors are simulated against the historical mean-covariance estimate; the highest-Sharpe portfolio found is retained. This provides a brute-force, assumption-light cross-check against the analytical optimizers below.
3. **Markowitz Mean-Variance Optimization** — classical efficient-frontier optimization (via PyPortfolioOpt's EfficientFrontier), solving for the maximum-Sharpe portfolio subject to long-only and per-position weight bounds.
4. **Black-Litterman** — combines market-cap-implied equilibrium returns ( $\pi = \delta \Sigma w_{\text{mkt}}$ , with  $\delta$  estimated from the SPY benchmark's implied risk aversion) with 50 explicit, hand-specified absolute return views, blended via Idzorek's method for view confidence weighting. This model is the most methodologically involved in the platform, and the one whose robustness was iteratively hardened during development (see Section 10).
5. **Risk-Parity** — allocates so that each asset contributes equally to total portfolio variance, rather than equalizing capital or targeting a mean-variance objective; included as a risk-based (rather than return-forecast-based) alternative.

**Shared constraints across all optimized models:**

| Parameter            | Value                                     |
|----------------------|-------------------------------------------|
| Weight bounds        | 0% – 30% per position (long-only, capped) |
| Risk-free rate       | 5.0% (annualized)                         |
| Covariance estimator | Ledoit-Wolf shrinkage                     |
| Lookback window      | 252 trading days (~1 year)                |

Each model's full-history ("in-sample") allocation is converted to a discrete share allocation for a \$1,000,000 initial capital base, so that reported weights correspond to an actually executable portfolio rather than a theoretical continuous allocation.

## 6. Risk & Performance Metrics

Every strategy is evaluated on a common metric set:

- Annualized return, annualized volatility, Sharpe ratio
- Sortino ratio (downside-deviation-adjusted)
- Calmar ratio (return relative to max drawdown)
- Maximum drawdown
- Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR)
- Win rate (share of positive-return trading days)
- Historical stress-period performance (see Section 9)

## 7. Transaction Cost & Execution Modeling

Frictionless backtests systematically overstate real-world performance, so the platform models trading costs explicitly as three additive components applied to rebalanced turnover:

| Cost component            | Rate                             |
|---------------------------|----------------------------------|
| Commission                | 0.05%                            |
| Bid-ask spread            | 0.03%                            |
| Market impact             | 0.02%                            |
| <b>Total assumed cost</b> | <b>0.1% per unit of turnover</b> |

The frictionless-market (in-sample) portfolio metrics for all five portfolio optimization models are summarized by the table below:

| Metric                                | Simulated (Max Sharpe Ratio) | Equal-weight | Markowitz | Black-Litterman | Risk-Parity |
|---------------------------------------|------------------------------|--------------|-----------|-----------------|-------------|
| Annualized Return                     | 16.6%                        | 15.5%        | 18.1%     | 11.7%           | 14.6%       |
| Annualized Volatility                 | 19.0%                        | 20.0%        | 17.8%     | 18.3%           | 17.7%       |
| Sharpe Ratio                          | 0.61                         | 0.52         | 0.73      | 0.37            | 0.54        |
| Maximum Drawdown                      | -45.7%                       | -46.6%       | -37.6%    | -43.0%          | -41.1%      |
| Sortino Ratio                         | 0.81                         | 0.69         | 0.98      | 0.47            | 0.71        |
| Calmar Ratio                          | 0.36                         | 0.33         | 0.48      | 0.40            | 0.35        |
| Value-at-Risk (VaR, 95%)              | -1.74%                       | -1.84%       | -1.58%    | -1.62%          | -1.63%      |
| Conditional Value-at-Risk (CVaR, 95%) | -2.75%                       | -2.89%       | -2.54%    | -2.67%          | -2.57%      |

On the other hand, the walk-forward-validated (out-of-sample) portfolio metrics for all five portfolio optimization models are summarized by the table below:

| Metric                                | Simulated (Max Sharpe Ratio) | Equal-weight | Markowitz | Black-Litterman | Risk-Parity |
|---------------------------------------|------------------------------|--------------|-----------|-----------------|-------------|
| Annualized Return                     | 24.1%                        | 20.8%        | 18.9%     | 18.6%           | 16.5%       |
| Annualized Volatility                 | 20.2%                        | 17.2%        | 24.6%     | 17.2%           | 14.1%       |
| Sharpe Ratio                          | 0.95                         | 0.92         | 0.56      | 0.79            | 0.81        |
| Maximum Drawdown                      | -21.3%                       | -23.9%       | -24.3%    | -24.6%          | -21.2%      |
| Sortino Ratio                         | 1.33                         | 1.36         | 0.798     | 1.12            | 1.19        |
| Calmar Ratio                          | 1.13                         | 0.869        | 0.777     | 0.755           | 0.776       |
| Value-at-Risk (VaR, 95%)              | -1.86%                       | -1.61%       | -2.17%    | -1.64%          | -1.32%      |
| Conditional Value-at-Risk (CVaR, 95%) | -2.89%                       | -2.33%       | -3.58%    | -2.41%          | -1.94%      |

Every model's performance is reported both **gross** (frictionless) and **net** (cost-adjusted) of these costs, both in-sample and out-of-sample, so the cost drag on each strategy's edge is directly visible rather than assumed away.

| Metric                     | Simulated (Max Sharpe Ratio) | Equal-weight | Markowitz | Black-Litterman | Risk-Parity |
|----------------------------|------------------------------|--------------|-----------|-----------------|-------------|
| Out-of-Sample Gross Return | 24.3%                        | 20.8%        | 19.1%     | 18.7%           | 16.5%       |
| Out-of-Sample Net Return   | 24.1%                        | 20.8%        | 18.9%     | 18.6%           | 16.5%       |
| Transaction-Cost Impact    | -0.2%                        | 0.0%         | -0.2%     | -0.1%           | -0.1%       |
| Daily Alpha                | 0.000140                     | 0.000199     | -0.000034 | 0.000163        | 0.000188    |
| Annualized Alpha           | 0.035879                     | 0.051472     | -0.008467 | 0.041952        | 0.048541    |
| R-squared                  | 0.763163                     | 0.855585     | 0.539304  | 0.640152        | 0.789515    |
| Residual Volatility        | 9.81%                        | 6.53%        | 16.71%    | 10.31%          | 6.49%       |
| Total Excess Return        | 20.1%                        | 17.2%        | 24.6%     | 17.2%           | 14.1%       |

## 8. Walk-Forward Validation Framework

This is the methodological core of the platform, and what distinguishes it from a single static backtest.

### 8.1 Rolling re-optimization (Phase 17)

At each scheduled rebalance date, all five models are **re-fit from scratch** using only the trailing 252-day window of data available *as of that date* — no future information is used at any point. Each model's re-optimization is wrapped in independent error handling, so a failure in one model on one date (e.g., insufficient market-cap history for Black-Litterman very early in the sample) does not halt the pipeline; it is logged as a diagnostic record and the pipeline proceeds. This produces a long panel of (date, model, asset) → weight allocations spanning the full 26.7-year history.

### 8.2 Out-of-sample backtest (Phase 18)

Using only rolling-window allocations that are available for **every** model on a given date (a data-quality gate added during development — see Section 10), the platform simulates a realistic walk-forward portfolio: allocations estimated at time  $t$  are held and only take effect from  $t + 1$  onward, are marked-to-market daily, and are rebalanced (with transaction costs deducted) whenever a new rolling allocation becomes available. This produces genuinely out-of-sample daily returns, turnover, and cost figures for every model — comparable directly against the in-sample, single-fit results from Section 5, so the "did re-optimization survive contact with new data" question can be answered with a number rather than assumed.

## 9. Parameter Robustness & Stress Testing (Phase 19)

A strategy that only works for one specific set of hyperparameters is not a strategy — it's an overfit backtest. Phase 19 systematically re-runs the walk-forward framework while varying:

- Lookback window length
- Maximum per-position weight cap
- Risk-free rate assumption
- Monte Carlo simulation sample size (for the Simulated model)

...and separately evaluates every model's performance across discrete historical stress periods (e.g., the 2008 financial crisis, the 2020 COVID drawdown, and other identifiable regime shifts within the sample), tracking each model's mean and **worst-case** Sharpe ratio across those periods — since a strategy's average performance across history is far less decision-relevant than how badly it degrades in the worst regime it has actually lived through.

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## 10. Performance Attribution & Risk Decomposition (Phase 20)

For the walk-forward (out-of-sample) results specifically, Phase 20 reconstructs each model's **daily asset-level return contributions** from its historical allocation panel, then:

- Runs factor-exposure regressions against momentum and volatility factors constructed endogenously from the platform's own 50-asset universe, to characterize *what kind of risk* each model's edge (or lack thereof) is actually exposed to.
- Decomposes total portfolio risk into asset-level contributions.
- Reconciles attributed returns back against the model's actual realized out-of-sample returns as an internal consistency check — ensuring the attribution math ties out rather than silently drifting from the true P&L.

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## 11. Software Architecture & Engineering Practices

- **Phased, sequential pipeline design** (20 numbered phases) with each phase's outputs explicitly validated before the next phase consumes them.
  - **Per-model, per-date error isolation**: optimization failures are caught, logged with diagnostic detail, and do not crash the run — appropriate for a 26-year, 5-model, rolling-window backtest where isolated data gaps (e.g., limited early-history share-count data for one name) are expected rather than exceptional.
  - **Explicit data-quality gating**: rebalance dates are only considered usable once verified to have allocations from every model, preventing silent gaps from propagating into the out-of-sample backtest as spurious errors.
  - **Compressed artifact output**: result tables are persisted as zstandard-compressed CSVs and figures are size-optimized via Pillow, keeping the run's output footprint manageable given the volume of daily, per-model, per-phase data produced.
  - **Consolidated reporting layer**: a final summary block programmatically identifies the top-performing model (by out-of-sample Sharpe) and prints its in-sample vs. out-of-sample performance, gross-vs-net transaction cost impact, factor exposure, and stress-test robustness in one place — so headline results don't require manually cross-referencing multiple output tables.
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## 12. Known Limitations & Data Considerations

- **Early-history data sparsity:** market-cap history (dependent on yfinance share-count data) is materially less reliable in the earliest years of the sample; the Black-Litterman model in particular cannot produce a valid allocation on rebalance dates before sufficient share-count history exists. This is handled (not hidden) via the model-coverage gating described in Section 8.2, rather than by silently interpolating cap data that doesn't exist.
  - **Discrete allocation rounding:** converting continuous optimal weights into whole-share positions on a fixed capital base introduces small, unavoidable tracking error versus the theoretical weights, more noticeable for higher-priced, lower-weight names.
  - **Transaction cost model is a flat-rate approximation:** the 10 bps assumption is a reasonable institutional planning estimate but does not model bid-ask spread widening under stress, partial fills, or borrow costs for any short exposure (the platform is long-only, so the latter does not apply).
  - **Universe survivorship:** the 50-name universe is fixed for the full backtest period; it does not model historical index additions/removals, which is a standard simplification but a real source of optimistic bias in any multi-decade backtest.
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## 13. Reproducing the Results

The script requires live internet access (for yfinance data retrieval) and the following key dependencies: pandas, numpy, scipy, PyPortfolioOpt, yfinance, zstandard, Pillow, and standard plotting libraries. Running the script end-to-end executes all 20 phases sequentially and concludes with a summary block reporting the winning model, its in-sample vs. out-of-sample Sharpe/return, transaction-cost impact, and stress-test robustness in a single consolidated printout.

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## 14. Conclusion

This platform's value is less in any single number it produces and more in its methodological discipline: every model is subjected to the same no-lookahead rolling re-estimation, the same realistic transaction costs, and the same out-of-sample validation before being declared a "winner." That discipline is what separates a credible quantitative research process from a backtest that merely looks good because it was fit to the data it was tested on — and it's the reason the platform reports gross-vs-net, in-sample-vs-out-of-sample, and mean-vs-worst-case figures side by side rather than a single headline metric.